Arithmetic landscapes III:
deformed measures, random sequences, and the coset identity

(subtitle provisional until the Part-III assembly is complete)

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Abstract

Papers 1–3 of this series fixed both the sequence and the measure on its subsets. Here we move each. Along the way the per-element energy $-\log|\cos \pi t|$ turns out to satisfy an exact coarse-graining identity: its average over a coset of index v is the same function at v times the frequency, plus $(1-1/v)\log 2$. We prove it twice, and it is what the minor-arc argument of this paper runs on. Replacing the uniform measure by a Bernoulli(q) product measure deforms the gap series into an explicit family $\Gamma^{(q)}$, and we prove that the fair coin minimises it for every sequence, with equality only at $q = \frac{1}{2}$; the argument is termwise and is machine-checked in Lean 4. We derive the corresponding transfer function $\Phi^{(q)}$ and show that the bias splits paper 3's [4] symmetric pair of strata in the ratio $q : (1-q)$, so that the biased landscape responds to a shift of target at first order; the resulting slope is a point prediction with no free parameter, and it is confirmed against the natural alternative to within 0.4%. Replacing the primes by a random odd sequence, we prove that the extremal modulus of the minor arcs moves from 6 to 4, so that the peak height available to a random odd sequence is $1/\sqrt{2}$ where the primes have $\sqrt{3}/2$ — the primes are the harder case. We then prove that the geometric *rate* follows: almost surely the minor-arc maximum of $F_A^{1/b}$ tends to $1/\sqrt{2}$. The proof carries no ineffective constant and needs no equidistribution input, because the coarse-graining identity replaces the usual discrepancy estimate by an equality. Two questions inherited from paper 3 are also settled or sharpened. Every claim is tagged with its status, and the numerical work is reproducible from logged scripts.

1 Introduction

Papers 1–3 fixed two things at once: the sequence A, and the measure on its subsets. Every statement of the trilogy is about a specific A under the uniform measure. This paper moves each of them, one at a time, and the two moves turn out to be worth making for different reasons.

Bias. The classification theorem of Part I [2] says which elements lie on which side of a target. It says nothing about how configurations are weighted, so the entire combinatorial layer survives verbatim when the uniform measure is replaced by the Bernoulli(q) product measure, and the gap series deforms into an explicit one-parameter family $\Gamma^{(q)}$ (§3). The answer has a shape worth putting in the introduction: **the fair coin gives the least rugged landscape**, for every sequence, by a termwise convexity argument that is machine-checked (Theorem 3.5, Remark 3.4). The transfer function deforms too, and here the deformation is

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not cosmetic: paper 3's cosh splits into two unequal exponentials with offsets in the ratio $q : (1-q)$, so the biased landscape responds to a shift of target at *first* order rather than second (§4). Rigidity off the fair coin is a tilt, not a plateau.

Coarse-graining. §2 is one lemma and it is the technical spine of the paper. Averaging the energy $X(t) = -\log|\cos \pi t|$ over the coset $\{t + k/v\}$ returns X at v times the frequency, plus $(1 - 1/v)\log 2$, exactly (Lemma 2.1, proved by the multiplication formula and again by Fourier). Since $X \geq 0$, the rational points are therefore the *minima* of the coset average, not merely convenient sample points (Corollary 2.2) — so the step of a minor-arc argument that normally costs a quantitative equidistribution estimate costs nothing here. Read as physics it is a block-spin step with an exact fixed point; we say so once, in Remark 2.4, and deduce nothing from the reading.

Randomness. For a random odd sequence the minor-arc input that cost Part II [3] fifty rounds is replaced by a concentration bound, and §5 carries *no ineffective constant at all* — the first such statement in the series. What we did not expect is that the random case is also *easier*: the extremal modulus moves from 6 to 4 (Theorem 5.5), so the peak height drops from $\sqrt{3}/2$ to $1/\sqrt{2}$, and the geometric rate improves in step (Theorem 5.8). The modulus moves for a one-line arithmetic reason — a set of primes contains at most one element $\equiv 3 \pmod 6$, a random odd set contains a third of them — and the consequence is that Part II's $\sqrt{3}/2$ is a property of the primes, not a limitation of the method. *The primes are the hard case.*

A methodological remark that this paper earned twice. Deformations are not only a way to generalise; they are a way to *see*. **A deformation that breaks a symmetry makes measurable everything the symmetry was hiding.** Constructing $\Gamma^{(q)}$ and $\Phi^{(q)}$ audited paper 3 for free, twice. First, the c_d fork: asking how the omitted factor behaves as k grows, rather than how large it is at one k, closed a question paper 3 had left open (§6). Second, and more sharply, the sign of λ: paper 3's Φ is an *even* function of λ, so no observable in that paper depends on the sign, and a definition that carried the wrong one survived three papers and an external referee. Splitting the cosh made the sign observable in a single measurement. We record the principle because it is a design rule, not an anecdote: when a quantity cannot be checked, deform the problem until it can.

Verification, and the genre. Every numbered claim carries a status tag: proved, Lean-verified, experimentally confirmed with the range stated, or conjectured. The scripts that produce every number in this paper write logs beside themselves and are part of the artefact. This is now a recognisable genre — machine-assisted mathematics shipped with formal certificates, of which *Ten advances in mathematics and theoretical computer science* [1] and its accompanying Lean 4 repository are the visible precedent. The direction differs: that work attacks named open problems, whereas the present series builds a small theory from its definitions upward, so the certificates here anchor *definitions and structural lemmas* rather than headline resolutions. The verification ethos is the same one, and we take it as read rather than arguing for it.

2 An exact coarse-graining identity

Papers 1–3 borrowed the vocabulary of statistical physics — landscape, energy, transfer function, tilt — and used it as a source of the right questions. This section is the first place where the borrowing runs the other way and produces an exact recursion rather than an analogy. Everything later in the paper that concerns minor arcs (§5.1, §5.2) is a corollary of it.

Throughout write

$$X(t) \;=\; -\log|\cos\pi t| \;\in [0,\infty],$$

the per-element energy of papers 1–3 at phase t.

Lemma 2.1 (coset identity)**.** *For every integer $v \geq 1$ and every real t,*

$$\frac{1}{v}\sum_{k=0}^{v-1} X\Big(t+\frac{k}{v}\Big) \;=\; \Big(1-\frac{1}{v}\Big)\log 2 \;+\; \frac{1}{v}X(vt+\tau_v), \qquad \tau_v = \begin{cases} 0, & v \text{ odd},\\ \frac{1}{2}, & v \text{ even}.\end{cases}$$

Equivalently $\prod_{k<v}|\cos\pi(t+k/v)| = 2^{1-v}|\cos\pi(vt+\tau_v)|$.

First proof (multiplication formula). Write $2\cos\pi z = 2\sin\pi(z+\frac{1}{2})$ and apply $\prod_{k<v} 2\sin\pi(y+\frac{k}{v}) = 2\sin(v\pi y)$ — the factorisation of $x^v - 1$ read on the unit circle — with $y = t+\frac{1}{2}$. The product of the $2|\cos|$ is $2|\sin\pi(vt+\frac{v}{2})|$, which is $2|\cos\pi vt|$ for v odd and $2|\sin\pi vt| = 2|\cos\pi(vt+\frac{1}{2})|$ for v even. Divide by 2^v and take logarithms. □

Second proof (Fourier). From $-\log|2\sin\pi u| = \sum_{n\geq 1}\cos(2\pi nu)/n$ and $u = t+\frac{1}{2}$,

$$X(t) = \log 2 + \sum_{n\geq 1}\frac{(-1)^n\cos(2\pi nt)}{n}.$$

Averaging over the coset annihilates every harmonic but those with $v \mid n$, since $v^{-1}\sum_{k<v}\cos(2\pi n(t+k/v))$ is $\cos(2\pi nt)$ when $v \mid n$ and 0 otherwise. Writing $n = vp$ and using $(-1)^{vp} = (-1)^p$ for v odd and 1 for v even, the surviving series is $v^{-1}(X(vt+\tau_v) - \log 2)$. □

Two proofs check the *statement*; one proof checks only the argument. The identity is verified in product form for $v \leq 60$ at 200 random shifts each, worst discrepancy $2.5\cdot 10^{-16}$, and the two Fourier partial sums agree to $8.8\cdot 10^{-14}$. [proved, twice; and checked numerically, `stab_r110`, `rate_r112`]

Corollary 2.2 (the rational points are the minima)**.** $\frac{1}{v}\sum_{k<v} X(t+\frac{k}{v}) \;\geq\; \Big(1-\frac{1}{v}\Big)\log 2$ *for every t, with equality exactly when $vt+\tau_v \in \mathbb{Z}$.*

Proof. $X \geq 0$. □

Remark 2.3 (what is machine-checked in Lemma 2.1, and what is not)*.* The identity is *multiplicative in v*, and that is what has been formalised. Writing $k = i + aj$ and inducting on b splits a product over $\{k < ab\}$ into a double product with no bijection and no analysis — `prod_range_mul_index`, stated for an arbitrary commutative monoid. On top of it, `cosetId_one` and `cosetId_two` are the base cases ($v = 2$ is the double-angle formula and needs nothing else), and `cosetId_mul` carries the identity from a and b to ab, including the offset bookkeeping $a\tau_b + \tau_a \equiv \tau_{ab} \pmod 1$ in all four parity cases, via `abs_cos_add_int_mul_pi`. *What is not formalised is the base case at an odd prime*, which is the only place a product over roots of unity is needed, and which Mathlib does not supply. So the machine-checked statement is: *the identity holds at* 1 *and* 2 *and is closed under products*, hence holds for every v built from the primes at which it is established by hand. The reduction was checked numerically before it was formalised, including a deliberately wrong τ as a control. [Lean-verified (the reduction; not the odd-prime base case); `Pnp/Theory/CosetProd.lean`, `coset_mult_r116`]

Remark 2.4 (what this says, and in which language)*.* The identity has two readings and they have different statuses; the paper keeps them apart.

As arithmetic (proved) it says that averaging X over a coset of index v returns X *itself, at v times the frequency*, plus a constant — so the family $\{X(v\,\cdot)\}$ is closed under coarse-graining, exactly, with no error term. That is what Corollary 2.2 turns into a uniform lower bound, and it is why §5.2 needs no equidistribution input: the question "how far can the mean fall when the rational grid is shifted?" has the answer *it cannot fall at all.*

As physics (an interpretation, and nothing is deduced from it) this is a block-spin step. Coarse-graining the phase lattice by a factor v maps the energy to a rescaled copy of itself; the map has an exact fixed-point structure rather than an approximate one, and the constant $(1-1/v)\log 2$ is the free energy released per coarse-graining step, additive over successive steps because τ composes. Papers 1–3 treated the landscape picture as a source of questions; this is the first identity in the series that the picture predicts and the arithmetic confirms. §7 says what we would want to do with it and does not do it here.

3 The biased invariant

Let P_q be the product measure under which each $a_i \in A$ is present independently with probability $q \in (0,1)$, and write $\mathrm{lm}_q(n)$, $\deg_q(n)$ for the P_q-measures of the sets of strict local minima and of ground states at the target n. The target is taken at the q-tilted centre $n_q = \lfloor qT \rceil$.

Definition 3.1 (biased gap series)**.**

$$\Gamma^{(q)}(A) \;=\; 1 + \sum_{d \geq 1} \Big[q^{N_d} + (1-q)^{N_d} \Big].$$

At $q = \frac{1}{2}$ every bracket is $2 \cdot 2^{-N_d}$ and $\Gamma^{(1/2)}$ is the window series W_D of Part I [2], hence $\Gamma(A) + (2D+1)2^{-k}$.

Remark 3.2 (why the classification survives the deformation)*.* Part I's classification is measure-free: at error d, an overshooting configuration is a strict local minimum exactly when every $a \leq 2d$ is *excluded*, and an undershooting one exactly when every $a \leq 2d$ is *included*. Under P_q those two events have probability $(1-q)^{N_d}$ and q^{N_d}, which is Definition 3.1. Writing $r_B^q(m) = \sum_{U \subseteq B,\ \sigma(U)=m} q^{|U|}(1-q)^{|B|-|U|}$ and factorising P_q across I_d and B_d,

$$\mathrm{over}_q = (1-q)^{N_d} r^q_{B_d}(n+d), \quad \mathrm{under}_q = q^{N_d} r^q_{B_d}(n-d-\sigma_d), \quad \deg_q = \sum_{J \subseteq I_d} q^{|J|}(1-q)^{N_d-|J|} r^q_{B_d}(n-\sigma(J)),$$

and the binomial sum over J is exactly 1, so a flat r^q gives the d-th term of Definition 3.1. [derived; the assignment verified per stratum, see Remark 3.8]

3.1 The fair coin is the flattest

Lemma 3.3 (termwise minimality)**.** *For every integer $N \geq 0$ the function $f_N(q) = q^N + (1-q)^N$ on $(0,1)$ is symmetric about $q = \frac{1}{2}$ and convex, and $f_N(q) \geq f_N(\frac{1}{2}) = 2^{1-N}$, with strict inequality for $q \neq \frac{1}{2}$ as soon as $N \geq 2$.*

Proof. $f_N(1-q) = f_N(q)$ is immediate. For $N \geq 2$, $f_N'(q) = N\big[q^{N-1} - (1-q)^{N-1}\big]$ and $f_N''(q) = N(N-1)\big[q^{N-2} + (1-q)^{N-2}\big] > 0$, so f_N is strictly convex; and $t \mapsto t^{N-1}$ is strictly increasing, so $f_N' < 0$ on $(0, \frac{1}{2})$ and $f_N' > 0$ on $(\frac{1}{2}, 1)$. For $N \in \{0,1\}$, f_N is the constant 2 or 1. $\square$

Remark 3.4 (formally verified). Lemma 3.3 is machine-checked. `termwise_min` gives the inequality for every N and every $q \in [0,1]$, `termwise_min_strict` its strictness for $N \geq 2$ and $q \neq \frac{1}{2}$, and `termwise_min_iff` the equality case as a biconditional; the normalisation $2 \cdot (\frac{1}{2})^N = 2^{1-N}$ is `two_mul_half_pow`. The non-strict bound is proved twice, once from convexity of x^N on $[0, \infty)$ and once from Bernoulli's inequality (`termwise_min'`), so the two proofs check the *statement* against each other and not merely the tactic script. No `sorry`, and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`. [Lean-verified; `Pnp/Theory/TermwiseMin.lean`]

Theorem 3.5 (the fair coin minimises the invariant)**.** *For every finite increasing sequence A of odd positive integers with $|A| \geq 2$ and every $q \in (0,1)$,*

$$\Gamma^{(q)}(A) \;\geq\; \Gamma^{(1/2)}(A),$$

with equality if and only if $q = \frac{1}{2}$.

Proof. Sum Lemma 3.3 over d. For strictness it suffices that some $d \leq D$ has $N_d \geq 2$: take any $d \geq (a_2 - 1)/2$, which is at most $D = (a_k - 1)/2$ because $a_2 \leq a_k$. □

Remark 3.6 (what the theorem says). $q(1-q)$ is the per-element variance of the coin. Biasing the coin in either direction lowers that variance and raises the invariant: **the more predictable the coin, the more rugged the landscape**. The effect is exactly symmetric in $q \leftrightarrow 1-q$, as it must be, since complementing every configuration exchanges q with $1-q$ and n with $T-n$.

Corollary 3.7 (closed form for the odd numbers)**.** *For $A = (1, 3, \ldots, 2k-1)$ one has $N_d = d$ for $d \leq k$, so in the limit $k \to \infty$*

$$\Gamma^{(q)} = 1 + \frac{q}{1-q} + \frac{1-q}{q} = \frac{1}{q(1-q)} - 1 \;\geq\; 3,$$

with equality only at $q = \frac{1}{2}$: the invariant of the odd numbers is the reciprocal of the coin's variance, minus one. *[verified against the summed series to $5.3 \cdot 10^{-15}$ over $q = 0.10, \ldots, 0.80$; e5_r098]*

3.2 Verification

[experimentally confirmed] Measured $\mathrm{lm}_q/\deg_q$ at n_q against the point prediction of Definition 3.1, relative error:

profile	q	$k = 60$	$k = 100$	$k = 140$
odds	0.20	$1.00 \cdot 10^{-2}$	$3.16 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$
odds	0.30	$2.63 \cdot 10^{-3}$	$8.02 \cdot 10^{-4}$	$3.75 \cdot 10^{-4}$
odds	0.50	$2.73 \cdot 10^{-4}$	$5.99 \cdot 10^{-5}$	$2.19 \cdot 10^{-5}$
primes	0.30	$2.18 \cdot 10^{-3}$	$5.02 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$
primes	0.50	$1.73 \cdot 10^{-4}$	$2.26 \cdot 10^{-5}$	$7.32 \cdot 10^{-6}$

There are no free parameters. Two features: the error decays faster than $1/k$ (fitted exponent 2.25 at $q = 0.2$ to 3.0 at $q = 0.5$), and it grows monotonically as q leaves $\frac{1}{2}$, by a factor ≈ 40 from $q = 0.5$ to $q = 0.2$ at fixed k. The second is what the biased flatness hypothesis will have to quantify. [**TODO** state the biased (H); design after §3.3.]

Remark 3.8 (the assignment, and a test that could not decide it). Definition 3.1 is symmetric in $q \leftrightarrow 1-q$, and so is the exact complementation identity $\mathrm{lm}_q(n) = \mathrm{lm}_{1-q}(T-n)$ — at the q-tilted centre, $T - qT = (1-q)T$. Neither can therefore distinguish the assignment of q^{N_d} and $(1-q)^{N_d}$ to the two strata: measured at $k = 100$, the ratio at $q = 0.4$ and at $q = 0.6$ agrees to every printed digit. The discriminating test is per stratum, $\mathrm{over}_q/\deg_q$ against $(1-q)^{N_d}$ and $\mathrm{under}_q/\deg_q$ against q^{N_d}; measured, both ratios lie in $[0.9978, 1.0018]$ over $d \le 6$, whereas a swap would disagree by a factor of 161 at $d = 6$. [experimentally confirmed; `e5_r098`]

3.3 The bias breaks the reflection symmetry

[experimentally confirmed, shape] Off the q-tilted centre the deviation $\mathrm{dev}_q = (\mathrm{lm}_q/\deg_q)/\Gamma^{(q)} - 1$ behaves differently at $q = \frac{1}{2}$ and away from it, and the reason is structural. Paper 3's transfer function is even in λ because the two strata sit at $\pm\delta_d$ about a common centre — the analytic form of $\mathrm{lm}(n) = \mathrm{lm}(T-n)$. Under bias that symmetry maps q to $1-q$ rather than to itself, so the odd orders need not cancel, and the first-order coefficient is

$$L^{(q)}(A) \;=\; \sum_{d\ge 1} \delta_d \Big[q^{N_d} - (1-q)^{N_d}\Big],$$

which is antisymmetric in $q \leftrightarrow 1-q$ and vanishes identically at $q = \frac{1}{2}$. Measured for the odd numbers at $k = 80$: $L^{(q)} = -28.634, -8.843, 0, +8.843, +28.634$ at $q = 0.3, 0.4, 0.5, 0.6, 0.7$, and for the primes -83.717, -23.677, 0, $+23.677$, $+83.717$.

The data agree. Over a range of targets, the spread (max over min) of $|\mathrm{dev}_q - \mathrm{dev}_q(0)|$ divided by λ_q and by λ_q^2:

profile	q	divided by λ_q	divided by λ_q^2
odds	0.30	**1.055**	3.778
odds	0.50	3.118	**1.002**
primes	0.30	**1.162**	4.149
primes	0.50	3.104	**1.012**

So the response is linear in λ under bias and quadratic at $q = \frac{1}{2}$, where $(\mathrm{dev} - \mathrm{dev}_0)/\lambda^2$ is constant to three digits — 19.8221 to 19.8586 for the odd numbers and 48.8625 to 49.4422 for the primes, across a threefold range of the target offset.

Remark 3.9 (the range is part of the test). The first measurement used offsets $x \in [0.04, 0.20]$ and came out incoherent for the primes at $q = \frac{1}{2}$ (a quadratic spread of 15.4). The reason is worth recording because it is a property of ratio tests and not of this problem. At $q = \frac{1}{2}$ the response is *even* in x, so $\mathrm{dev} - \mathrm{dev}_0$ cannot change sign for the reason under test; at finite k the vertex of the parabola sits at some $x_* \neq 0$, and $\mathrm{dev} - \mathrm{dev}_0$ crosses zero at $\pm x_*$. Measured, $x_* \approx 0.03$, with the signs symmetric in $\pm x$ exactly as that account predicts. A range containing the crossing makes *both* ratios pass through zero, so the spread of either is set by how close a grid point landed to x_* — an accident of the grid. **A ratio test must not straddle a zero of its own numerator.** The table above uses $x \in [0.10, 0.30]$, outside x_*; the repair is the range, not the omission of the offending point.

[experimentally confirmed; `e6_r100`, `e6b_r115`, whose pipeline is checked against `e6_r100` by reproducing $[19.8186, 19.8352]$ on the original range, and whose discriminating control at $q = 0.3$ still returns *linear* on the new range]

4 The biased transfer function

4.1 The sign of λ, which now matters

Paper 3's transfer function is even in λ, so the sign of the slope is not observable there. Splitting the cosh makes it observable, so we fix it first:

$$\lambda_q(n) \;=\; +\frac{d}{dm}\log r_A^q(m)\Big|_{m=n} \;=\; \tfrac14\log\frac{r_A^q(n+2)}{r_A^q(n-2)}+O(\cdot),$$

positive below the biased centre. This is the convention under which $\lambda \approx s$; paper 3's `def:slope` carried the opposite sign and has been corrected, with no numerical consequence there for exactly the reason just given.

4.2 The derivation

Take $P_{q,s}$: element a present with weight $q\,e^{-sa}$, absent with weight $1-q$, so $p_a = qe^{-sa}/((1-q)+qe^{-sa})$ and $p_a = q-q(1-q)sa+O(s^2a^2)$. Two consequences, both absent at $q=\frac12$:

- the per-element variance is $a^2p_a(1-p_a) = q(1-q)a^2(1+O(sa))$, so the layer weight of paper 3's `rem:mushift` becomes $e^{-\lambda^2 q(1-q)s_d/2}$;
- removing I_d removes mean $q\sigma_d$, so the two strata no longer sit at $\pm\delta_d$ about a common centre. They sit at

$$\delta_d^+ = d+q\sigma_d, \qquad -\delta_d^-, \quad \delta_d^- = d+(1-q)\sigma_d.$$

The bias splits the pair asymmetrically, in the ratio $q:(1-q)$.

Hence

$$\Phi^{(q)}(\lambda) \;=\; 1+\sum_{d\ge1}\Big[(1-q)^{N_d}e^{\lambda\delta_d^+}+q^{N_d}e^{-\lambda\delta_d^-}\Big]e^{-\lambda^2q(1-q)s_d/2}. \tag{1}$$

[derived; three internal checks below are exact, the point prediction is verified]

Three checks are built in, and all three hold to machine zero ($0.00\cdot10^0$, two profiles, $k=60$): $\Phi^{(q)}(0)=\Gamma^{(q)}$ of Definition 3.1; the complementation symmetry $\Phi^{(q)}(\lambda)=\Phi^{(1-q)}(-\lambda)$, which holds because $\delta^+(q)=\delta^-(1-q)$; and $\Phi^{(1/2)}$ is paper 3's Φ term by term. [experimentally confirmed; e10_r105]

4.3 Tilted rigidity

Proposition 4.1 (the biased landscape responds at first order)**.** $\Phi^{(q)\prime}(0)=\sum_{d\ge1}\Big[(1-q)^{N_d}\delta_d^+-q^{N_d}\delta_d^-\Big]$, *which vanishes identically at $q=\frac12$ and is non-zero otherwise. So rigidity at a biased measure is not flatness: it is a* tilt *of a definite slope, and the slope is a point prediction with no free parameter.*

[derived and verified, not proved from a hypothesis-free statement: it is a consequence of $\Phi^{(q)}$, itself derived by the partition-function argument of §4 and confirmed by three exact internal checks. The first-order coefficient is a point prediction with no free parameter, and §4.4 tests it against the naive alternative on 36 discriminating rows — which is why the two are distinguished here rather than only in §8.]

This is what §3.3 demanded of the derivation, and it is a sharper demand than it looks. The naive guess — keep paper 3's $\delta_d = d+\sigma_d/2$ and merely split the weights — gives instead $\sum[(1-q)^{N_d}-q^{N_d}]\delta_d$, and the two differ by exactly $(\frac12-q)\sum_d\sigma_d[q^{N_d}+(1-q)^{N_d}]$: zero at $q=\frac12$, and 20–32% of the prediction at the q tested. The measurement therefore decides between them.

4.4 Verification

The estimator must kill constants, because the point prediction of §3.2 carries an $O(10^{-3})$ offset at the biased centre which is the same size as the linear term. So we use a central difference about n_q: with $n_\pm = \lfloor (q \pm hq)T \rceil$,

$$\widehat{L}(h) = \frac{(\mathrm{lm}_q/\deg_q)(n_+) - (\mathrm{lm}_q/\deg_q)(n_-)}{\lambda_q(n_+) - \lambda_q(n_-)}.$$

Ratios of $\widehat{L}$ to the two candidates, at $h = 0.01$:

profile	q	separation	$\widehat{L}/\Phi^{(q)\prime}(0)$	$\widehat{L}$/naive
odd numbers, $k = 120$	0.30	31.6%	**0.9945**	0.6805
odd numbers, $k = 120$	0.40	19.9%	**0.9976**	0.7991
odd numbers, $k = 120$	0.50	—	$\Delta(\mathrm{lm}_q/\deg_q) = 0$ exactly	
odd numbers, $k = 120$	0.60	19.9%	**0.9976**	0.7991
primes, $k = 120$	0.30	32.3%	**0.9981**	0.6754
primes, $k = 120$	0.40	20.2%	**0.9983**	0.7967

Over all 36 discriminating rows (two profiles, $k = 80, 120$, four q, four h), the median $|\widehat{L}/\Phi^{(q)\prime}(0) - 1|$ is 0.0041 against 0.2038 for the naive candidate. The one row above 10% is the primes at $k = 80$, $q = 0.3$, $h = 0.01$, where $\Delta\lambda = 8.5 \cdot 10^{-5}$ and the integer rounding of $n_\pm$ is a visible fraction of the step; the same row at $k = 120$ reads 0.9981. At $q = \frac{1}{2}$ the numerator is 0 or $4.4 \cdot 10^{-16}$, below the stated floor — the fair coin has no linear response, which is the content of Proposition 4.1 at its zero. [experimentally confirmed; `e10_r105`]

The split is also visible stratum by stratum, which tests the offsets directly rather than through their sum: at $k = 80$, $q = 0.3$, $\mathrm{over}_q/\deg_q$ against $(1-q)^{N_d} e^{\lambda\delta_d^+} e^{-\lambda^2 q(1-q)s_d/2}$ agrees to within 0.24% over $d = 1, \ldots, 6$, and $\mathrm{under}_q/\deg_q$ against its partner to within 0.9%. [experimentally confirmed; `e10_r105`]

4.5 The biased hypothesis

(H_q): $\sum_d [(1-q)^{N_d}(\delta_d^+)^2 + q^{N_d}(\delta_d^-)^2] < \infty$, uniformly for $q \in [q_0, 1-q_0]$. Since $\max(q, 1-q) \leq 1 - q_0 < 1$, the geometric damping of paper 3's (H) survives with base $1 - q_0$, and the constant degrades as $q_0 \to 0$ — which is the $\sim 40\times$ growth of the §3.2 error already measured between $q = \frac{1}{2}$ and $q = 0.2$. The $\alpha = 1$ threshold of paper 3 is unchanged, the damping argument being base-agnostic. [derived; the degradation matches the measured trend]

5 Random sequences, unconditionally

Before the theorem, the floor it has to stand on. Let A be drawn from the Cramér model on the odd integers: each odd $m \in [3, N]$ is included independently with probability $2/\log m$, and N is fixed by $\mathbb{E}|A| = k$. Nothing is conditioned, so $|A|$ fluctuates from instance to instance. That is deliberate: $kR_A(k)$ is the k-invariant quantity of paper 3's `rem:onek`, so an ensemble of genuinely different sizes tests the $1/k$ law for free. Twenty instances at each of two sizes, each through the whole paper-3 pipeline with its own N_d and σ_d:

$\mathbb{E}\|A\|$	range of k	(H) series	mean kR	sd	sd/mean
120	101–135	133.79–320.79	214.96	10.58	4.92%
180	146–191	132.94–471.77	206.32	6.94	3.37%

at $x = 0.06$; $c_A = kR/100$. Three things, and the third is the one to keep. [experimentally confirmed; e7_r102]

(i) The hypothesis holds on the ensemble. All forty instances have a bounded (H) series and a window that tracks the primes'. This is a precondition, not a result: a theorem about random sequences is unconditional only if the random sequences satisfy (H) themselves.

(ii) The constant concentrates. The spread across instances is under 5% and falls with k, while the mean of kR moves by 4% between the two sizes — the $1/k$ law of paper 3 surviving an ensemble it was not fitted on.

(iii) The primes sit inside the ensemble and the odd numbers do not. Against the $k \approx 180$ ensemble (2.0632 ± 0.0694 in c_A), the primes score $z = -0.39$ and the odd numbers $z = -2.70$, at every one of the four targets. In this statistic the primes are indistinguishable from a random odd sequence of the same density; the odd numbers are a different object.

There is one place where the random model is *not* like the primes, and it is the place that matters for the minor arcs. Write $h_d(q) = \max_{(r,q)=1} \big(\prod_{a \in B_d} |\cos(\pi r a/q)|\big)^{1/|B_d|}$. Since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a, the modulus-4 floor $h_d(4) = 1/\sqrt{2}$ is deterministic, not merely universal. Part II [3]'s vanishing lemma kills the modulus-$2m$ peak (m odd) as soon as B_d contains an element $\equiv 3 \pmod 6$. A set of primes contains exactly one such element, namely 3, and 3 leaves B_d at $d \geq 2$, so the primes keep $h_d(6) = \sqrt{3}/2$; a random odd sequence contains about $|B_d|/3$ of them and loses the peak at every d. Measured over $q \leq 120$ at $d = 1, 2, 5$: every random instance has $\max_{q \geq 3} h_d(q) = 1/\sqrt{2}$ attained at $q = 4$, and the primes have $\sqrt{3}/2$ at $q = 6$. [experimentally confirmed; e7_r102]
So the random theorem should come out with the geometric rate $1/\sqrt{2} = 0.7071$ per element rather than the $\sqrt{3}/2 = 0.8660$ of Parts I and II [2, 3]: *the primes are the harder case, and the modulus-6 ripple is a structural feature of the primes that the random model destroys.*

5.1 The modulus-4 theorem

The observation above is not an accident of the ensemble; it is a theorem about residues, and it is the exact counterpart of Part II's modulus-6 theorem for a different equidistribution class. Part II averages $|\cos|$ over the *reduced* residues, which is the right class for the primes. A random odd sequence is equidistributed over the *odd* residues instead, so the relevant quantity is a different one.

Definition 5.1. For $q \geq 2$ let $R_q = \mathbb{Z}/q$ if q is odd and $R_q = \{j : j \text{ odd}\} \subset \mathbb{Z}/q$ if q is even — for q even and m odd, $m \bmod q$ is odd, so R_q is the set of residues an odd sequence can occupy. Put

$$M_{\mathrm{odd}}(q) \;=\; \Big(\prod_{j \in R_q} |\cos(\pi j/q)|\Big)^{1/|R_q|}.$$

For r coprime to q the map $j \mapsto jr$ permutes R_q, so $M_{\mathrm{odd}}(q)$ does not depend on the numerator of the rational r/q.

Everything below is a corollary of the coset identity of §2, because $\{j/q : j \in R_q\}$ *is* a coset of $(1/v)\mathbb{Z}$.

Corollary 5.2 (the identity in the form §5.2 uses)**.** *Let $q \geq 2$, put $v = q$ for q odd and $v = q/2$ for q even, and let $\sigma = \frac{1}{2}$ if $q \equiv 2 \pmod 4$ and $\sigma = 0$ otherwise. Then $|R_q| = v$ and, for every real s,*

$$\prod_{j \in R_q} \Big|\cos \pi\Big(s + \frac{j}{q}\Big)\Big| \;=\; 2^{1-v}\, |\cos \pi(vs + \sigma)|, \qquad \textit{hence} \qquad \frac{1}{v} \sum_{j \in R_q} X\Big(s + \frac{j}{q}\Big) \;\geq\; \Big(1 - \frac{1}{v}\Big) \log 2.$$

Proof. $\{j/q : j \in R_q\}$ is $\{k/v\}_{k<v}$ for q odd and $\{\frac{1}{2v} + \frac{k}{v}\}_{k<v}$ for q even. Apply Lemma 2.1 at $t = s$ and at $t = s + \frac{1}{2v}$ respectively. In the second case the shift contributes $v \cdot \frac{1}{2v} = \frac{1}{2}$ to the argument, so the total offset is $\tau_v + \frac{1}{2}$, which is an integer when v is even and $\frac{1}{2}$ when v is odd; and v odd with q even is exactly $q \equiv 2 \pmod 4$. The inequality is Corollary 2.2. □

Lemma 5.3 (closed form)**.** *$M_{\mathrm{odd}}(q) = 2^{1/q-1}$ for q odd. For $q = 2u$ even, $M_{\mathrm{odd}}(q) = 2^{1/u-1}$ if u is even, and $M_{\mathrm{odd}}(q) = 0$ if u is odd.*

Proof. Put $s = 0$ in Corollary 5.2: the right-hand side is 2^{1-v} when $\sigma = 0$ and 0 when $\sigma = \frac{1}{2}$. Take v-th roots. □

Remark 5.4 (a second proof of the odd case, with no roots of unity). For q odd the closed form also follows from the double-angle formula alone, and this is the route we formalise. Let $P = \prod_{j=1}^{q-1} |\sin(\pi j/q)|$ and $C = \prod_{j=1}^{q-1} |\cos(\pi j/q)|$. From $\sin 2x = 2 \sin x \cos x$, $\prod_{j=1}^{q-1} |\sin(2\pi j/q)| = 2^{q-1} PC$. Since q is odd, $j \mapsto 2j \bmod q$ is a bijection of $\{1, \ldots, q-1\}$ and $|\sin(2\pi j/q)| = |\sin(\pi\,(2j \bmod q)/q)|$, so the left side is P again. As $P \neq 0$, $C = 2^{1-q}$, and $\prod_{j \in R_q} |\cos(\pi j/q)| = C$ because the $j = 0$ factor is 1. Two proofs of one load-bearing identity check the *statement*; one proof checks only the argument. This is the branch that is machine-checked, as `prod_abs_cos_odd`; see Remark 5.6.

Theorem 5.5 (the modulus-4 theorem)**.** $\max\limits_{q \geq 2} M_{\mathrm{odd}}(q) = \dfrac{1}{\sqrt{2}}$*, attained at $q = 4$ and nowhere else.*

Proof. $v \mapsto 2^{1/v-1}$ is strictly decreasing. For q odd, $q \geq 3$, Lemma 5.3 gives $M_{\mathrm{odd}}(q) = 2^{1/q-1} \leq 2^{-2/3}$. For $q \equiv 2 \pmod 4$ it gives 0. For $q \equiv 0 \pmod 4$, write $q = 2u$ with u even, $u \geq 2$; then $M_{\mathrm{odd}}(q) = 2^{1/u-1} \leq 2^{-1/2}$ with equality exactly when $u = 2$, i.e. $q = 4$. And $2^{-2/3} < 2^{-1/2}$. □

Lemma 5.3 is verified against the product for all $q \leq 200$, worst discrepancy $4.4 \cdot 10^{-16}$; Corollary 5.2 is verified in its product form for $q \leq 120$ at 400 random shifts each, worst discrepancy $4.4 \cdot 10^{-16}$, and independently at 60 decimal digits, worst discrepancy $2.2 \cdot 10^{-61}$. [proved; and checked numerically, `mq4_r107`, `stab_r110`]

Remark 5.6 (what is machine-checked here, and what is not). The two halves of the argument have different statuses and the difference is worth stating. *The maximisation is formally verified.* Taking the closed form of Lemma 5.3 as a definition, `ModdCF_le` gives $M_{\mathrm{odd}}(q) \leq 2^{-1/2}$ for every $q \geq 2$, `ModdCF_eq_iff` the equality case as a biconditional, `ModdCF_lt_of_ne_four` the strict bound off $q = 4$ by a second route, and `two_rpow_neg_half` the normalisation $2^{-1/2} = 1/\sqrt{2}$. This is the step where an error would hide — a case split over three residue classes and the comparison of $2^{-2/3}$ with $2^{-1/2}$ — and it is now checked by machine. *Of the closed form itself, the odd branch is now formalised and the even branch is not.* `prod_abs_cos_odd` proves $\prod_{j=1}^{q-1} |\cos(\pi j/q)| = (1/2)^{q-1}$ for every odd q, by the double-angle route of Remark 5.4 — which is exactly why that route is worth having: it uses one bijection and one trigonometric identity, where the proof of Lemma 2.1 needs a product over roots of unity that Mathlib does not carry. The even branch, and Lemma 2.1 in general, remain proved on paper and checked numerically. [Lean-verified (maximisation, and the odd closed form); `Pnp/Theory/ModFour.lean`, `Pnp/Theory/OddProd.lean`]

Remark 5.7 (the two classes, side by side).

	class of residues	extremal q	rate per element
primes (Part II)	reduced	6	$\sqrt{3}/2 = 0.8660$
random odd A (here)	odd	4	$1/\sqrt{2} = 0.7071$

The two disagree exactly at $q \equiv 2 \pmod 4$: $M(6) = \sqrt{3}/2$ but $M_{\mathrm{odd}}(6) = 0$, because $j = 3$ is an odd residue mod 6 and $\cos(\pi/2) = 0$. A set of primes contains at most one element $\equiv 3 \pmod 6$, and it leaves B_d at $d \geq 2$; a random odd sequence contains a positive proportion of them at every d. *So the primes are the harder case, by a factor $\sqrt{3/2} = 1.2247$ in the geometric rate*, and Part II's $\sqrt{3}/2$ is a feature of the primes rather than a limitation of the method.

Both halves of that comparison are machine-checked. `abs_cos_pi_mul_div_four` gives $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a, so `prod_abs_cos_four` makes the modulus-4 floor $(1/\sqrt{2})^{|A|}$ *deterministic* rather than merely universal; and `cos_pi_mul_div_six_eq_zero` gives $\cos(\pi a/6) = 0$ whenever $a \equiv 3 \pmod 6$, so `prod_abs_cos_six_eq_zero` kills the modulus-6 peak on a single element. `six_dies_four_survives` is the separation itself: for an odd sequence containing such an element the modulus-6 product is 0 while the modulus-4 product is still exactly $(1/\sqrt{2})^{|A|} > 0$. [Lean-verified; `Pnp/Theory/OddPeaks.lean`]

5.2 The rate, and what is still missing

Theorem 5.8 (minor-arc rate for random odd sequences)**.** *Let A be drawn from the Cramér model above and let $F_A(\theta) = \prod_{a \in A} |\cos(\pi a \theta)|$. Then almost surely*

$$\lim_{b \to \infty} \max_{\theta \text{ minor}} F_A(\theta)^{1/b} = \frac{1}{\sqrt{2}}.$$

Here minor *means $\|\theta\| \geq 1/N$, with $\|\cdot\|$ the distance to the nearest integer; Step 2$''$ shows that this radius is large enough, so nothing is left unowned.*

[proved, with every constant written out. The proof is the four steps below rather than a proof environment, so it is worth saying here that it is one: no step is deferred, no constant is fitted, and the only external input is the arithmetic half, Theorem 5.5, which is proved in this paper. See Remark 5.10 for what an independent reader should check first.]

The arithmetic half is Theorem 5.5. Here is the rest, in the steps it takes, with the constants named. Fix the Dirichlet parameter $Q = \lfloor \sqrt{N} \rfloor$ — Step 2$'$ says why that value and not another — and set

$$M = \log \frac{2Q}{\pi} + 1, \qquad \boxed{\delta = \tfrac{1}{6} \log 2 = 0.1155245\ldots}$$

Write $X_m(\theta) = -\log|\cos(\pi m \theta)| \in [0, \infty]$, so that $-\log F_A(\theta) = \sum_m \xi_m X_m(\theta)$ with ξ_m the independent indicators of the model. Only a *lower* bound on this sum is wanted, which is what makes the argument easy.

Step 1 (truncate; the unbounded tail is free). Put $X_m^{(M)} = \min(X_m, M)$. Then $\sum \xi_m X_m \geq \sum \xi_m X_m^{(M)}$, and the summands now lie in $[0, M]$. At $\theta = \frac{1}{4}$ truncation is inactive, since $X_m(\frac{1}{4}) = \frac{1}{2} \log 2$ for every odd m.

Step 2 (the mean, and the constant δ). At $\theta = r/q$ the values X_m over odd m run through $\{-\log|\cos(\pi j/q)| : j \in R_q\}$, so by Lemma 5.3 their mean is $-\log M_{\mathrm{odd}}(q) = (1 - 1/v) \log 2$ and the *gap* over the value $\frac{1}{2} \log 2$ at $q = 4$ is

$$\delta(q) = \left(\tfrac{1}{2} - \tfrac{1}{v}\right) \log 2, \qquad v = q \ (q \text{ odd}), \quad v = q/2 \ (q \equiv 0 \bmod 4), \quad \delta = \infty \ (q \equiv 2 \bmod 4).$$

This vanishes exactly at $v = 2$, i.e. $q = 4$, and increases with v; the smallest admissible v after 2 is $v = 3$. **So the binding competitor is $q = 3$ and the gap is $\delta = \frac{1}{6} \log 2$,** with $q = 8$ next at $\frac{1}{4} \log 2$. Step 2$'$ will replace the entry $\delta = \infty$ at $q \equiv 2 \pmod 4$ by the finite value $(\frac{1}{2} - \frac{1}{u}) \log 2$: off the rational point the vanishing residue no longer vanishes, and $q = 6$ then *ties* with $q = 3$ at $\frac{1}{6} \log 2$. The constant is unchanged and two moduli bind it rather than one.

The two moduli with $v \le 2$ — $q = 4$, where the floor is $\frac{1}{2}\log 2$ exactly and is meant to be, and $q = 2$, where $v = 1$ and the floor degenerates to 0 — are handled in Step $2''$, together with $q = 1$. [proved from Lemma 5.3; checked against the direct residue average for all $q \le 400$, worst discrepancy $1.0 \cdot 10^{-15}$, and the minimum located at $q = 3$ over $q \le 2000$; `delta_r109`]

Truncation does not eat this. For $q \not\equiv 2 \pmod 4$ one has $|\cos(\pi j/q)| \ge \sin(\pi/2q)$, so $\max_j X_m \le \log(2q/\pi) < M$ and truncation is *inactive* at every $q \le Q$ — measured, the largest value over all such $q \le 1000$ is 6.4552 against $M - 1 = 6.4562$, so the bound $\log(2Q/\pi)$ is sharp and the $+1$ in M is there to keep the argument off a knife edge. For $q = 2u \equiv 2 \pmod 4$ truncation *is* active, at the single odd residue $j = u$ where the cosine vanishes; putting $x = -1$ in $(x^u + 1)/(x + 1)$ gives $\prod_{j \ne u} |\cos(\pi j/q)| = u\, 2^{1-u}$, so the truncated mean is $[(u-1)\log 2 - \log u + M]/u$, which exceeds $\frac{1}{2}\log 2 + \delta$ by between 0.23 and 1.25 over $q \le 198$ at $Q = 200$. [experimentally confirmed, exact to $4.4 \cdot 10^{-16}$; `delta_r109`]

Step 2′ (from the rationals to every θ: there is no dent). Write $\theta = a/q + \beta$ with $q \le Q$ and $|\beta| \le 1/(q(Q+1))$ (Dirichlet). The obstacle here looks like a job for a quantitative equidistribution estimate — Erdős–Turán or Koksma — and it is not: Lemma 2.1 settles it as an *identity*. Since $j \mapsto ja$ permutes R_q, the values $\{ja/q\}$ are the $\{j/q\}$, and for any common shift s

$$\frac{1}{|R_q|}\sum_{j \in R_q} X\Big(s + \frac{j}{q}\Big) \;=\; \underbrace{\Big(1 - \frac{1}{v}\Big)\log 2}_{-\log M_{\mathrm{odd}}(q)} \;+\; \frac{1}{v}\,X(vs + \sigma) \;\ge\; -\log M_{\mathrm{odd}}(q), \tag{$*$}$$

because $X \ge 0$. *Shifting the whole rational grid can only raise the mean*; the dent one expects when θ moves off a rational does not exist, and no discrepancy theory is needed to bound it. Equality in $(*)$ holds exactly on $vs + \sigma \in \mathbb{Z}$, so the rational values are the true minima, not merely convenient sample points. [proved (Lemma 2.1); the minimum over s of the left side of $(*)$ is checked to equal $-\log M_{\mathrm{odd}}(q)$ to $3 \cdot 10^{-16}$ for 15 moduli, and to survive truncation at M for every $q \le 1000$; `stab_r110`]

First, truncation does not spoil $(*)$. The loss $(X - M)^+$ is nonzero only within e^{-M}/π of a pole, and the coset points are $1/v \ge 1/Q > 1/(Qe) = 2e^{-M}/\pi$ apart, so at most one residue is affected; if it lies at distance d from the pole then $X \le \log\frac{1}{2d}$, while the excess term of $(*)$ is $\frac{1}{v}X(vs+\sigma) \ge \frac{1}{v}\log\frac{1}{\pi v d}$ because $vs + \sigma$ sits at distance vd from a pole. The floor therefore survives as soon as $M \ge \log(\pi v/2)$, and with $M = \log(2Q/\pi) + 1$ and $v \le Q$ that reads $1 \ge \log(\pi^2/4)$ — true, **with a margin of exactly** $1 - \log(\pi^2/4) = 0.0968$. This is the third place the $+1$ in M earns itself. [proved; and $\min_s \Phi_q^{(M)}(s) - (1 - 1/v)\log 2$ is checked to be $\ge -5.6 \cdot 10^{-16}$ for every $q \le 1000$; `stab_r110`, `rate_r112`]

Step $2'$ continued (the two error terms, in full). What $(*)$ does not by itself cover is that element m of the class j sits at $ja/q + m\beta$, so the shift varies *within* a class. Fix $q \notin \{1, 2\}$ and group the odd $m \in [3, N]$ by residue class mod q. In a class the m form an arithmetic progression of common difference $2v$ (for q odd the odd numbers in a class mod q step by $2q = 2v$; for $q = 2u$ they step by $q = 2v$), so the phases form an arithmetic progression of step $\eta = 2v\beta$ sweeping an arc of length $L = N|\beta|$, and the class starts at $x_j + m_j\beta$ where $m_j \le 2v$ is its first element. Three errors, and nothing else:

(i) Progression versus integral. $X^{(M)}$ is of bounded variation, at most $2M$ per period, so for a progression of step η over an arc of length L, $|\sum_r X^{(M)} - \eta^{-1}\int| \le 2M(L+1)$. There are v classes and $\asymp N/2$ terms in all, so in the mean this costs at most $4qM/N + 4M/(Q+1)$, using $v|\beta| \le 1/(Q+1)$.

(ii) Jitter. Replacing the arc of class j by the common arc is a translation by $\epsilon = m_j\beta$, and $m_j \le 2v$ while the progression step is $\eta = 2v\beta$, so $|\epsilon| \le |\eta|$ *exactly*. Translating by ϵ changes the integral by at most ϵV, so the per-class error is at most $\epsilon V/\eta \le V \le 2M(L+1)$ — the same shape as (i), and not a separate mechanism at all. Per element, $4qM/N + 4M/(Q+1)$.

(iii) Unequal class counts. They differ by at most 1, costing vM in the sum, i.e. $2qM/N$ in the mean.

After (i)–(iii) the integrand is the honest coset average, which is $\geq (1-1/v)\log 2$ pointwise by $(*)$; integrating a pointwise inequality needs no further hypothesis, and in particular *no case distinction on the length of the sweep* — which is what makes the route robust in Q where an arc-comparison design is not. Collecting,

$$\frac{\sum_m p_m X_m^{(M)}(\theta)}{\sum_m p_m} \;\geq\; \Big(1-\frac{1}{v}\Big)\log 2 \;-\; E, \qquad E \;\leq\; \frac{10\,qM}{N}+\frac{8M}{Q+1}.$$

Both terms are minimised together at $Q=\lfloor\sqrt{N}\rfloor$, where $E=O(M/\sqrt{N})$ — which is why the Dirichlet parameter is $\sqrt{N}$ and not, as one first writes, N. At $Q=N$ the argument fails twice over: the residue classes hold $O(1)$ elements each, and $|\beta|\leq 1/(qN)$ makes every sweep shorter than a single period.

The weights need no argument at all. $p_m=2/\log m$ is *decreasing*, so by Abel summation a bound valid for every prefix $\{m\leq N'\}$ passes to the p-weighted mean unchanged: if $S(N')\geq c\,P(N')-\mathcal{E}(N')$ for all N', then $\sum p_m X_m^{(M)} \geq c\sum p_m - \sum_k (p_k-p_{k+1})\mathcal{E}(k) - p_N\mathcal{E}(N)$, and with $\mathcal{E}(N')\leq CM(\sqrt{N}+N'/\sqrt{N})$ the subtracted term is $O(M\sqrt{N})+O(M/\sqrt{N})\sum p_m$, i.e. $O(M\log N/\sqrt{N})$ relative. There is no block-weight condition to check. [proved; the shortfall at the binding modulus $q=3$ is measured as $1.9\cdot 10^{-3}, 5.8\cdot 10^{-4}, 1.7\cdot 10^{-4}, 4.8\cdot 10^{-5}$ at $N=2001, 8001, 32001, 128001$, and deficit $\cdot\, N/(qM)$ is $0.3057, 0.3140, 0.3199$ at $N=8001, 32001, 128001$ — constant, i.e. the measured deficit has exactly the derived shape qM/N. Errors (i), (ii), (iii) are additionally isolated and measured *one at a time* against their own bounds, over three sizes, eight moduli and three offsets, worst values $1.1\cdot 10^{-2}$, $1.2\cdot 10^{-2}$ and $8.5\cdot 10^{-4}$ against bounds of order 10^{-1}; `stab_r110`, `rate_r112`, `audit_r114`]

Step 2″ (the three moduli with $v\leq 2$, and the major arc). Corollary 2.2 gives $\frac{1}{2}\log 2$ at $q=4$ ($v=2$) and nothing at all at $q\in\{1,2\}$ ($v=1$). The last two are not prose; they are the following lemma, and $q=4$ needs no more than the identity because $\frac{1}{2}\log 2$ is exactly the value the theorem is about.

Lemma 5.9 (the degenerate moduli)**.** *Let $\phi(w)=-\log|\sin\pi w|$ and $L>0$. Then $\frac{1}{L}\int_0^L\phi = \log 2+\mathrm{Cl}_2(2\pi\{L\})/(2\pi L)$, and*

$$\frac{1}{L}\int_0^L \phi \;>\; \tfrac{1}{2}\log 2 \quad \textit{for every } L>0, \qquad \frac{1}{L}\int_0^L X \;\geq\; \log 2-\frac{\mathrm{Cl}_2(\pi/3)}{2\pi} = 0.5316 \quad \textit{for } L\geq 1.$$

Consequently the mean over a $q=2$ sweep exceeds $\frac{1}{2}\log 2$, and so does the mean over a $q=1$ sweep once $L=N\|\theta\|\geq 1$ — which is exactly the definition of minor*.*

Proof. The closed form is $\int_0^x\phi = x\log 2+\mathrm{Cl}_2(2\pi x)/(2\pi)$ together with the periodicity of ϕ. For $L\leq\frac{1}{2}$: ϕ is convex on $(0,1)$, so the average over $[0,L]$ is at least the midpoint value $-\log\sin(\pi L/2)\geq -\log\sin(\pi/4)=\frac{1}{2}\log 2$; at the endpoint $L=\frac{1}{2}$ the two-point midpoint bound $\frac{1}{2}[\phi(L/4)+\phi(3L/4)] = 0.51986$ shows the inequality is strict there. For $L\geq\frac{1}{2}$: $1/(2\pi L)\leq 1/\pi$ and $|\mathrm{Cl}_2|\leq \mathrm{Cl}_2(\pi/3)$, so the average is at least $\log 2-\mathrm{Cl}_2(\pi/3)/\pi = 0.3701 > \frac{1}{2}\log 2$. For $q=1$ the same computation with X in place of ϕ gives $\log 2+\mathrm{Cl}_2(2\pi L+\pi)/(2\pi L)$, and $L\geq 1$ makes the correction at most $\mathrm{Cl}_2(\pi/3)/(2\pi)=0.1615$. □

The true $q=2$ floor is 0.4945 at $L=0.7908$ — above $\frac{1}{2}\log 2+\delta$, not merely above $\frac{1}{2}\log 2$ — but that is measured, not proved, and the theorem does not need it. The smallest L at which the $q=1$ average reaches $\frac{1}{2}\log 2+\delta$ is 0.4528, so the radius $1/N$ in the definition of *minor* is comfortable rather than tight. [proved; the Clausen closed form checked against direct quadrature to $4.5\cdot 10^{-5}$, and the convexity bound verified at 5000 values of L; `rate_r112`]

Step 3 (concentration). The summands are independent and lie in $[0, M]$, and there are $n = \lfloor (N-1)/2 \rfloor$ of them, so Hoeffding gives $\Pr[\sum \xi_m X_m^{(M)} < \mathbb{E} - t] \le \exp(-2t^2/(nM^2))$. Take $t = \varepsilon\,\mathbb{E}|A|$ and use $\mathbb{E}|A| \asymp 2N/\log N$, $n \asymp N/2$, $Q = \lfloor \sqrt{N} \rfloor$ and hence $M \asymp \frac{1}{2}\log N$: the failure probability at one θ is

$$\exp(-c\,\varepsilon^2 N/\log^4 N), \qquad c = 64.$$

Two readings of the same bound, and the theorem uses the second. With $\varepsilon = \frac{1}{2}\delta$ this is the fixed-margin statement, $\exp(-cN/\log^4 N)$ with $c = 16\delta^2 = 0.2135$. But the theorem only needs the mean to survive to within $o(1)$, so one may instead take $\varepsilon = N^{-1/4}$, which still leaves the exponent $\asymp \sqrt{N}/\log^4 N \to \infty$. *That is why $q \in \{1, 2\}$ need only clear $\frac{1}{2}\log 2$ and not $\frac{1}{2}\log 2 + \delta$*, and it is what makes Lemma 5.9 sufficient. No second moment of an exponential sum appears: the randomness is in the *sequence*, not in the phases.

Step 4 (net, then Borel–Cantelli). Where $X_m < M$ we have $|\cos(\pi m\theta)| \ge e^{-M}$, hence $|\partial_\theta X_m^{(M)}| = \pi m |\tan(\pi m\theta)| \le \pi N e^M \le 2eNQ \le 2eN^{3/2}$, so a net of spacing $\varrho/(2eN^{3/2}b)$ suffices and has $O(N^{5/2}/\varrho)$ points. Since $N^{5/2}\exp(-c\varepsilon^2 N/\log^4 N)$ is summable in N for $\varepsilon = N^{-1/4}$, the union bound and Borel–Cantelli give the almost-sure statement.

Assembling. Steps 1–4 give, almost surely and uniformly over minor θ, $b^{-1}\sum_{a\in A} X_a(\theta) \ge \frac{1}{2}\log 2 - o(1)$, i.e. $\max_{\theta \text{ minor}} F_A(\theta)^{1/b} \le 2^{-1/2}(1 + o(1))$: at $q = 4$ the floor is $\frac{1}{2}\log 2$ by Corollary 2.2, at every other $q \ge 3$ it is $\frac{1}{2}\log 2 + \delta$, at $q \in \{1, 2\}$ it is above $\frac{1}{2}\log 2$ by Lemma 5.9, and the error of Step 2′ is $O(M\log N/\sqrt{N})$. The matching lower bound is free: $\theta = \frac{1}{4}$ is minor and $X_a(\frac{1}{4}) = \frac{1}{2}\log 2$ for *every* odd a, so $F_A(\frac{1}{4})^{1/b} = 2^{-1/2}$ identically. Hence the limit is $2^{-1/2}$. □[proved]

Remark 5.10 (what "proved" means here, and what it does not). Theorem 5.8 carries no hypothesis and every constant in its proof is written down. One qualification, stated here rather than left for a referee to raise. The three error terms of Step 2′ are each reduced to a standard estimate — a Riemann-sum comparison for a function of bounded variation, and Abel summation against a monotone weight — which we state and apply in the form we need but do not prove; a referee would be within rights to ask for them in full, and doing so would lengthen Step 2′ without changing any constant. *No hypothesis is not the same as written out in full*, a distinction Part II now also makes explicitly about its own main theorem, and the two should be read the same way.

Measured, at $\theta = r/q$ for $q \le 12$ and eight instances at $b \approx 190$: the mean of $b^{-1}\sum_a -\log|\cos(\pi a/q)|$ agrees with $-\log M_{\mathrm{odd}}(q)$ within the $O(b^{-1/2})$ fluctuation at every q, and at $q = 4$ it is 0.346574 with standard deviation 0.000000 — deterministic, as Definition 5.1 says it must be. [experimentally confirmed; `mq4_r107`, `delta_r109`]

Remark 5.11 ($\theta = \frac{1}{4}$ is the limit of the maxima, not a maximum). A first scan of $F_A(\theta)^{1/b}$ over a fine net asked whether anything exceeds $1/\sqrt{2}$, and found that something does. That is not a counterexample and the reason is worth stating, because it fixes the form of Theorem 5.8: the maximum is *approached* at $\frac{1}{4}$, and the statement is about the limit of $\max_\theta F_A^{1/b}$, not about where a maximum sits at finite b. Moving off $\frac{1}{4}$ by δ changes $\log F_A^{1/b}$ by $\delta\, b^{-1}\sum_a(\pm\pi a) + O(\delta^2)$, and the signs depend on $a \bmod 4$, so the first-order term is a random walk of size $b^{-1}\sqrt{S_2}$. Optimising δ gives an excess of $O(1/b)$ in the exponent — a *bounded multiplicative factor* on F_A, not a geometric one. Measured over $b = 111, 190, 310, 542$: $\max_\theta F_A^{1/b} - 1/\sqrt{2}$ is $9.0\cdot 10^{-4}$, $1.8\cdot 10^{-3}$, $1.1\cdot 10^{-4}$, $1.2\cdot 10^{-3}$, and $b\log(\sqrt{2}\max_\theta F_A^{1/b})$ stays in $[0.05, 0.89]$ — $O(1)$, as derived. The argmax sits at $\theta = 0.24996, \ldots, 0.25001$. [experimentally confirmed; `mq4_r107`]

6 Two questions inherited from paper 3

(i) The c_d fold-in is settled, and it goes paper 3's way. The share of the residual that c_d carries falls like $k^{-2\alpha}$ — measured exponents 1.98, 2.35, 2.99 against predicted 2.00, ≈2.4, 3.00 on the odd numbers, the primes and $\alpha = \frac{3}{2}$, with the squares already below the numerical floor. So Φ as printed is right for every larger k and c_d belongs to E_k permanently; paper 3's `rem:cd` now says so. [experimentally confirmed, four profiles, $k \leq 300$; `e8_r103`]

(ii) $c_A = c'_A$ is still open, but sharper. Carried to $k = 300$ the ratio reaches 1.0064 (odd numbers), 1.0001 (primes), 0.9984 ($\alpha = \frac{3}{2}$), and the gap closes like $k^{-2.4}$ — faster than the $1/k$ law both constants sit on. [experimentally confirmed; `e8_r103`] [**TODO** the mechanism is still missing, and a measured rate of convergence is evidence about a limit, not a proof of one. F45 stands: substituting s for λ makes the residual three times worse.]

7 Where this sits, and the algorithmic reading

7.1 Iterating the coarse-graining

Lemma 2.1 is used once, in one direction, and that is not the only thing it says. Written as an operator, $(\mathcal{C}_v f)(t) = v^{-1}\sum_{k<v} f(t+k/v)$ acts on X by $\mathcal{C}_v X = (1-1/v)\log 2 + v^{-1}X(v\,\cdot + \tau_v)$, so X spans a one-dimensional affine invariant subspace of every $\mathcal{C}_v$ at once, and the constants compose: $\mathcal{C}_{v'}\mathcal{C}_v = \mathcal{C}_{vv'}$ on this subspace, with released constants adding as $(1-1/v)+v^{-1}(1-1/v') = 1 - 1/(vv')$. Three questions we have not attempted, listed so that the omission is deliberate rather than silent. *(i)* Which energies other than X have this property — is $-\log|\cos|$ characterised by being an exact fixed point of the family $\{\mathcal{C}_v\}$ up to affine terms? *(ii)* Is there a transfer-operator reading in which Φ of paper 3 is the spectral object and $\mathcal{C}_v$ the renormalisation map? *(iii)* Papers 1–3's landscape language was a source of questions; here it produced an identity. Does the same happen for the *biased* energy of §4, where the coset structure is weighted by q? [conjectural directions; nothing in this paper depends on any of them]

7.2 Related work

Subset sums and number partitioning have a large landscape literature, and it is worth saying precisely where the present series does and does not meet it. That literature is overwhelmingly about *random instances*: the phase transition and barrier-tree statistics of the partitioning landscape, the local-REM description of the near-optimal spectrum, and more recently overlap-gap obstructions for local algorithms. Its objects are the energy spectrum and the dynamics; its randomness is in the instance; and its answers are laws for a typical instance.

Papers 1–3 do something different in kind: they fix the sequence and compute an *arithmetic invariant of that set*, $\Gamma(A) = 1 + 2\sum_d 2^{-\#\{a\in A:\, a\leq 2d\}}$, which the ratio $\mathrm{lm}/\deg$ converges to for *every* target. Nothing is averaged over instances, and the answer is a number one reads off A. The present paper's §5 is the one place where the two meet — there the sequence is random — and even there the question is the ratio's limit rather than the spectrum's law.

One vocabulary warning, since the words collide. "Biased Bernoulli measure" is established terminology in the theory of Bernoulli convolutions, where it denotes a self-similar measure on $\mathbb{R}$; that is a different object from P_q here, and we do not claim the term. Within the subset-sum landscape literature we have not found the biased deformation $\Gamma^{(q)}$ of §3, nor the odd-residue extremal quantity M_{odd} of §5.1, under any name. [literature pass, not a proof of novelty: absence of evidence over the searches made]

7.3 The algorithmic reading

This programme began with local search on subset-sum instances, and the invariant answers a question one asks there. If a local-search procedure ends at a uniformly chosen strict local minimum, the expected number of attempts before landing on a ground state is $\mathrm{lm}/\deg$, and papers 1–3 predict that this number converges to $\Gamma(A)$ — *computable from A alone, without solving a single instance, and independent of the target.* For the odd numbers it is 3; for the primes, 5.3492... Two conditions are attached to that sentence and we attach them here rather than leaving them in paper 3: the convergence is a *proof skeleton* there, not a theorem (paper 3, Theorem 3.1, whose missing ingredient is named in its own statement), and the uniformity of the terminal distribution is an assumption about the search, not a fact about the landscape.

Theorem 3.5 is the part of this subsection that has a proof, and it is a statement about *sampling*, not about search dynamics: drawing configurations with a biased coin instead of a fair one multiplies the ratio by $\Gamma^{(q)}/\Gamma \geq 1$, strictly for $q \neq \frac{1}{2}$. So a natural one-parameter family of sampling rules cannot beat the fair coin on this statistic — which is a claim about the statistic, and becomes a claim about a restart heuristic only under the uniformity assumption just named. Section 4 adds that the biased landscape also acquires a *first-order* sensitivity to the target that the fair one does not have.

Behind the uniformity assumption is the reason this is an outlook and not a theorem about algorithms: a count of local minima is not a distribution over basins of attraction, and if the basins are very unequal the ratio bounds the wrong quantity. Nor does anything here address dynamics — barrier heights, escape times, or the behaviour of any particular heuristic. What the invariant supplies is a target-independent constant that any such analysis has to be consistent with, together with a proved statement that one obvious way of trying to improve that constant does not improve it. [**TODO** if this section is ever to make an algorithmic claim, the missing object is the basin-size distribution, not more landscape statistics.]

8 Honest scope

- *Lean-verified*: Lemma 3.3 in all three forms (Remark 3.4), including the equality case; and the maximisation step of Theorem 5.5 (Remark 5.6), the closed form being taken as a definition there; and the two peak facts of Remark 5.7 (`Pnp/Theory/OddPeaks.lean`); and the odd branch of Lemma 5.3 itself, by the double-angle route (`Pnp/Theory/OddProd.lean`, Remark 5.4). The canon also passes an independent kernel replay (`lean4checker` replays every constant of the import closure of the root module `Pnp` from the imports up, with a negative control). Each file named above is inside that closure; a file present in the source tree but outside it is *not* replayed, and the harness now fails if one exists, so this sentence cannot quietly become false.

- *Proved*: Lemma 3.3, Theorem 3.5 and Corollary 3.7; Lemma 2.1 (twice) with Corollaries 2.2 and 5.2; Theorem 5.5 and Lemma 5.3; Lemma 5.9; and Theorem 5.8, all constants written out. Also Remark 3.2's derivation of Definition 3.1, modulo the biased flatness hypothesis, which is not yet stated.

- *Experimentally confirmed, with the range stated*: the point prediction (§3.2, two profiles, five q, three k), the stratum assignment (Remark 3.8), the linear response under bias (§3.3).

- *Derived and verified*: $\Phi^{(q)}$ of (1) and Proposition 4.1 — three exact internal checks, and the first-order point prediction confirmed against the naive alternative on 36 discriminating rows (§4.4); the random-ensemble floor of §5; the c_d verdict of §6.

- *Interpretation, with nothing deduced from it*: the renormalisation reading of Lemma 2.1 (Remark 2.4) and the three questions of §7.
- *Not yet written*: the passage from (H_q) to a rigidity statement.

Data availability

Every number in this paper is produced by a script that writes a log beside itself, as in papers 1–3. The scripts, the logs, the Lean development and the checks that enforce all of this are public at `https://github.com/mathlab0911/arithmetic-landscapes`.

References

[1] OpenAI, *Ten advances in mathematics and theoretical computer science*, 2026, with Lean 4 certificates at `github.com/openai/ten-proofs`. Cited here only as a precedent for the genre; no mathematical result of that work is used.

[2] K. Amauchi, *Arithmetic landscapes I: the gap series.* Manuscript, 2026.

[3] K. Amauchi, *Arithmetic landscapes II: asymptotic flatness of subset-sum landscapes of primes.* Manuscript, 2026.

[4] K. Amauchi, *The transfer function of subset-sum landscapes.* Manuscript, 2026. Being absorbed into the present Part; see the note on scope.